

Ang Zhe Hao (Nicholas)

+65 8508 0360 | nicholasang.2024@smu.edu.sg | <https://nicholassang.com>
<https://github.com/nicholassang/> | <https://www.linkedin.com/in/nicholas-ang-zhe-hao/>

EDUCATION

Singapore Management University

Aug 2024 – May 2028

- BSc (Software Engineering), second major in Software Systems
- Cumulative Grade Point Average (CGPA): 3.41/4.00
- Coursework: Cyber-Physical Systems, Advanced Databases, Generative AI with LLMs

TECHNICAL SKILLS

- Languages: Python, C++, Java, Typescript, Javascript, PHP, SQL, HTML/CSS
- Frameworks & Libraries: Node.js, Express.js, FastAPI, Spring Boot, React.js, Vue.js, Tailwind CSS
- Databases & Messaging: PostgreSQL, RabbitMQ, MySQL, DynamoDB
- DevOps & Tools: Docker, Docker Compose, AWS, Jira, Git
- Certifications: AWS Certified Solutions Architect - Associate, AWS Certified Developer - Associate

PERSONAL PROJECTS

Real-Time Multiplayer Card Game | Node.js, Express.js, WebSockets, AWS EC2, DynamoDB | [GitHub](#) | [Website](#)

- Engineered a server-authoritative real-time engine using Node.js and WebSockets to synchronize 2-player card movement, maintaining deterministic 20 FPS state consistency across concurrent game sessions
- Designed and executed a load test simulating 100 concurrent WebSocket players, sustaining 100% successful connections, ~100 messages/sec, and average connection latency 94ms, ensuring reliable real-time gameplay
- Implemented an O(1) in-memory matchmaking queue with RESTful DynamoDB APIs for match persistence, supporting scalable, low-latency matchmaking and long-term match storage

Global Movie Location Explorer | Python, FastAPI, Docker Compose, PostgreSQL, AWS Lightsail | [GitHub](#) | [Website](#)

- Engineered an automated ETL pipeline ingesting 300+ global filming locations in ~14 minutes with 96% data integrity through Selenium, automating IMDb data scraping, normalization, and loading for real-time analytics
- Optimized PostgreSQL queries with indexes on movie_id and lat/lng, achieving sub-3ms retrieval per movie and supporting efficient geospatial filtering across hundreds of locations
- Developed containerized backend services with Docker Compose and FastAPI endpoints, deployed to AWS Lightsail, enabling interactive mapping visualizations with heatmaps and toggleable point maps updated every 6 hours
- Implemented detailed logging at each ETL stage, enabling monitoring, debugging, and traceability of collected data

ACADEMIC PROJECTS & HACKATHONS

SchoolFinder Application | Coursework | Node.js, Express.js, Google Maps API, Google Cloud Platform, Vue.js | [GitHub](#)

- Developed a location-based web application rendering 500+ school markers with dynamic filtering and route computation, enabling users to visually explore schools across Singapore
- Reduced redundant API requests by >90% through backend caching, lowering cloud compute costs

Social Calendar Platform | SMU .Hack HEAPS Programme | Node.js, Express.js, PostgreSQL, React.js | [GitHub](#)

- Led a 5-person team over 12 weeks, delivering weekly sprint milestones for a multi-user calendar platform, ensuring timely project completion
- Designed a normalized PostgreSQL schema supporting multi-calendar event storage and cross-calendar filtering, enhancing database efficiency and scalability

Facial Recognition GUI | Coursework | Java, Springboot, OpenCV, ONNX (ArcFace), Apache POI | [GitHub](#)

- Developed a file export subsystem using Java and Apache POI, generating CSV, Excel, and PDF reports with student IDs, facial recognition metrics, and categorized report details
- Implemented structured logging and audit tracking, ensuring traceable, reliable reporting and compliance with usage tracking requirements

LEADERSHIP & VOLUNTEERING

President | SMU Strategica Board Games Club | 2026

- Directed 8-member executive committee managing weekly operations for 70+ active members
- Coordinated sponsorship outreach and cross-club partnerships to expand participation and event scale

Volunteer | SMU Project Kidleidoscope | 2024

- Conducted learning activities for primary school students